# Written System Description

Smart VMS (Smart Visitor Management System) — Government Systems Prototype Showcase and National Innovator Registry

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| **Version** | 1.0 |
| **Showcase repo** | https://github.com/paradigmaRnD/smartvms-showcase |

## 1. Executive Summary

Smart VMS is a production digital visitor management platform that replaces paper visitor logbooks at government and institutional reception points. The system registers visitors, records check-in and check-out events, notifies hosts, and provides searchable records and reports for security and compliance. Access is controlled by role, communications are encrypted, and each organisation's data is kept separate. The solution is deployed and in use, with a mobile channel for front-desk staff and a web channel for administrators.

## 2. Problem Statement

Paper visitor books remain common in ministries, agencies, hospitals, and offices. They are hard to search, offer weak privacy, provide limited audit evidence, and cannot alert hosts in real time. During security reviews or incidents, reconstructing who visited and when is slow and unreliable.

## 3. Solution Overview

Smart VMS delivers structured visitor workflows through a mobile application for reception and security teams and a web portal for organisation administrators. Tenants (organisations) manage their own premises, users, and policies. Supported workflows include walk-in visitors, scheduled visits, premises access validation, and administrative reporting—all subject to institutional access rules.

## 4. Key Capabilities

- End-to-end digitisation of the visitor logbook workflow with institutional multi-tenancy
- Configurable verification steps at check-in (including premises tokens and document-assisted capture)
- Policy-based access control across reception, security, host, and administrator roles
- Tamper-evident audit trail of visitor and administrative actions
- Operational resilience: offline-capable mobile use with secure synchronisation when online
- Integration-ready service layer for future Government identity and shared platforms

## 5. Target Users

- Government ministries and agencies (reception, security, protocol)
- Hospitals and health facilities
- Utilities and regulated enterprises
- Corporate and shared facilities

## 6. Deployment Status

Production deployment operational; mobile distribution via application stores; organisations onboarded and in active use.

## 7. Benefits to Government

- Replaces paper logbooks with searchable, consistent digital records
- Improves reception security with time-stamped check-in/out and access policies
- Supports audits and investigations through exportable activity logs
- Scales across multiple premises under one organisation
- Provides a foundation for future National ID and Government platform integration

## 8. Technical Summary (Declaration)

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| Languages | TypeScript, JavaScript, Dart |

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| **Frontend** | Flutter (mobile), React (web administration portal) |
| **Backend** | Node.js application services (modular API layer) |
| **Database** | Document-oriented primary store; relational store for administrative records |
| **Hosting** | Cloud-hosted production with HTTPS; container-ready deployment; on-premise option supported |
| **Mobile** | Yes |
| **Capacity** | 500+ per deployment instance; designed for horizontal scaling |

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